Experiment No. 16 – Sobel Algorithm Using OpenCV

1. Aim

To implement the Sobel algorithm using OpenCV to filter the input image and detect edges in horizontal and vertical directions.

2. Program

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

# Upload image
uploaded = files.upload()

# Read uploaded image
filename = list(uploaded.keys())[0]
image = cv2.imdecode(
    np.frombuffer(uploaded[filename], np.uint8),
    cv2.IMREAD_COLOR
)

if image is None:
    print("Error: Image not loaded")
else:
    # Convert image to grayscale
    gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
```

```
# Apply Sobel operator in X direction
sobel_x = cv2.Sobel(
    gray, cv2.CV_64F, 1, 0, ksize=3
)

# Apply Sobel operator in Y direction
sobel_y = cv2.Sobel(
    gray, cv2.CV_64F, 0, 1, ksize=3
)

# Convert results to absolute values
abs_x = cv2.convertScaleAbs(sobel_x)
abs_y = cv2.convertScaleAbs(sobel_y)

# Combine both directions
sobel_combined = cv2.addWeighted(
    abs_x, 0.5, abs_y, 0.5, 0
)

# Display results
plt.figure(figsize=(12, 8))

plt.subplot(2, 2, 1)
plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
plt.title("Original Image")
```

```
plt.axis("off")

plt.subplot(2, 2, 2)
plt.imshow(gray, cmap='gray')
plt.title("Grayscale Image")
plt.axis("off")

plt.subplot(2, 2, 3)
plt.imshow(abs_x, cmap='gray')
plt.title("Sobel X Direction")
plt.axis("off")

plt.subplot(2, 2, 4)
plt.imshow(sobel_combined, cmap='gray')
plt.title("Sobel Edge Detection")
plt.axis("off")

plt.tight_layout()
plt.show()
```

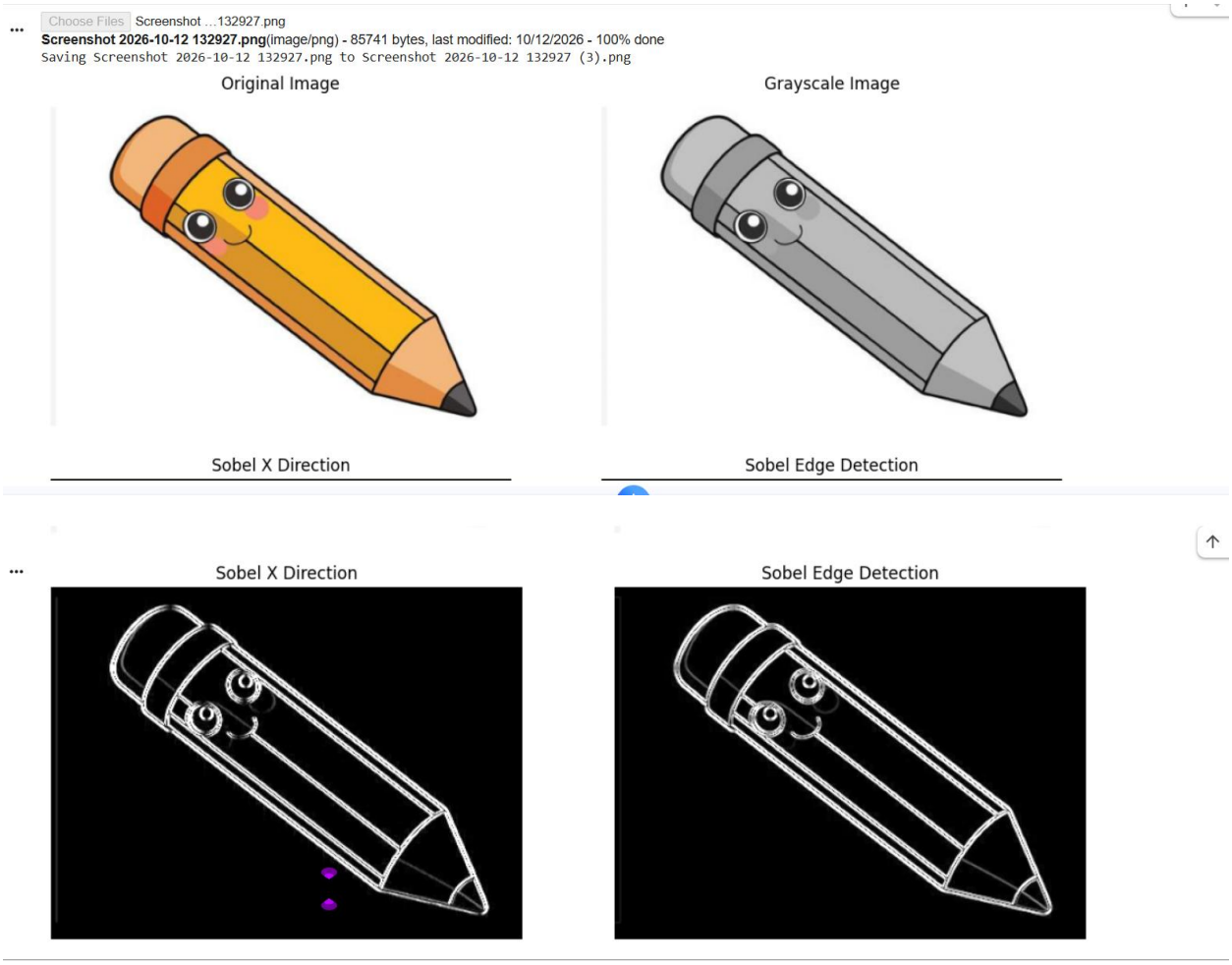
3. Expected Output

The output displays:

- Original Image
- Grayscale Image
- Sobel X Direction
- Sobel Edge Detection

Edges of the objects are highlighted in the processed image.

4. Output



The input image is successfully processed using the Sobel operator. Horizontal and vertical edges are detected and displayed.

5. Result

Thus, the Sobel algorithm was successfully implemented using OpenCV to filter the input image and detect its edges.